



BSB 513/BENG455- Systems Biology
Term Project

Students:

Merve Say 195023062008

Mustafa Şin 171134042

Lecturer: Dr. Tunahan Çakır

Gene Expression Analysis of Peripheral Blood Mononuclear Cells

1. Abstract

We performed gene expression analysis of peripheral blood mononuclear cells (PBMC) in patients with Peripheral arterial disease (PAD) and controls without PAD to identify significantly changed genes. For this analysis, using open source dataset from specific series accession number of the NCBI the Gene Expression Omnibus (GEO) database^[3]. This dataset is sorted using some packages in R.

We aim to get a result for molecular clues with determining associated the disease. For this reason, significant changes in genes are defined by using t-test and Benjamini and Hochberg False Discovery Rate (FDR) with 10% cutoff ($p < 0.1$). The number of these genes is 64. These genes are performed for enrichment analysis using Gene Ontology (GO) terms, pathway analysis using Kyoto Encyclopedia of Genes and Genomes (KEGG). For the top six most significant genes Gaussian Graphical Models (GGM) inference and the Pearson correlation methods applied to achieve the interaction between those genes.

2. Literature Information

Peripheral Arterial disease is the insufficient blood flow of artery in the peripheral regions of the body which means that away from the heart like legs, head, arms etc. The atherosclerosis is the buildup body metabolites like fats, cholesterol and some others in the artery plaque and it restrict the blood flow (ischaemia). In most cases atherosclerosis is charged as the cause of PAD but for many professionals it's underdiagnosed. In mild conditions PAD symptoms are pain in the lower extremities and discontinuous limping. Since the symptoms are not produced simultaneously from the beginning atherosclerosis, the number of patients is unknown. When it comes the acute progressing levels of the disease results could be gangrene amputation etc. In daily routine many risk factors such as smoking, physical inactivity, overweight, unhealthy diet (including high sugar and lipid) are responsible from PAD. There is no therapy that can be stated as the exact treatment but for many cases treatment basically reversing the above factors. Diagnose of the PAD is examination of physical activity, blood pressure in lower extremities with Doppler probe for basic diagnose, the MR, CT and angiography etc are used for an exact diagnosis.^[1,5]

Masud et.al^[1] was trying to identify significantly effective genes in case of the PAD formation. Their aim was earlier detection and prognostication of this disease with the help of the study inferences and outputs. The peripheral blood mononuclear circulating in the peripheral

vessel and has direct contact with the arterial wall. Therefore, PBMC are used in this study to measure differentially expressed genes in the formation of PAD.

Table-1 shows characteristics of the sample. The detailed information about recruitment and sample characteristics are given in the Masud et.al article and also isolation of blood cells and isolation of RNA process and procedures are given in this article. Microarray analysis were performed to measure the quantity and quality of the RNA. The produced gene expression data is available in Geo as “GSE27034”.^[1]

In this study whole dataset was used to achieve the significantly changed genes. Those genes enriched and analyzed in some databases and web tools to identify related biologically important processes, pathways or interactions etc. These results will give the information about how metabolic pathways are affected from PAD, which protein interactions occur during symptomatic PAD or which biological processes are responsible for the pain or any other symptoms. Detailed information will be given in the methods and results part.

Characteristics	Cases (n=19)	Controls (n=18)
Age, Years	69.57+-9.18	66.88 +- 9.44
Men	13(68%)	13(72%)
Ever smoker	17(89%)	12(66%)
BMI,kg-m2	28.53 +- 5.25	30.6 +-5.99
Systolic BP,mmHG	139.2 +- 23.27	132.4 +- 20.1
Diastolic Bp, mmHG	74.2 +- 11.66	80.1 +- 12.3
HDL cholesterol, mg/dl	44.12+- 8.59	49.7 +-15
Hypertension	17(89.4%)	13(72%)
Diabetes	5(26%)	4(22%)
Lipid lowering medication	11(57%)	12(6%)
Ankle brachial index	0.48 +- 0.20	1.12 +-0.07
WBC count	8.03 +- 2.34	7.26 +- 1.10

Table-1: Characteristics of the sample

3. Method

3.1 Sample and Dataset

We performed a comparison based on several gene expression datasets publicly available in the NCBI the GEO database under series accession number GSE22034. This dataset is associated with microarray studies. This platform is used with Human Genome U133 Plus 2.0 arrays from Affymetrix, containing 54675 probe sets. There are 37 samples of peripheral blood mononuclear cells (PBMC) with PAD and without PAD. We analyzed two different groups: 19 samples for Control Group and 18 samples for PAD Group.

3.2 Work Packages and Statistical Analysis

R programming language are used to make statistical analysis and plotting. To get the data set we use the "GEOquery" package. To draw plots or graphs ("ggplot2") and ("ggfortify") packages are used. Control and patient group separated. Principal component analysis (PCA) was performed ("prcomp") and the results did not seem to be correct or useful. Since that, we used an unpaired t-test and then we calculated the p-value. 36 genes calculated as significantly changed. Then Benjamini-Hochberg correction method are used with cut-off (FDR) "0.1". Based on those genes hierarchical clustering performed. In clustering the "pearson" method used for similarity in both genes and samples. Centroid linkage used to cluster samples and complete linkage method used to cluster genes. Dendrogram and heatmap graphs plotted to see the outlier sample. PCA analysis performed again and related graphs plotted to see outlier samples. Both clustering and PCA graphs are used to double check the outlier sample. After we determine the outlier sample it subtracted from data to get more accurate results.

3.3 GO and Pathway Enrichment Analysis

GO terms enrichment and pathway analysis with KEGG was performed using g:GOST that is one of the g:Profiler tools. For identified biological processes and pathway analysis, we upload our significantly changed gene list on this platform.

3.4 Gaussian graphical models-based network inference

Top 6 most significantly changed genes were used to make GGM based network inference and Pearson correlation. Results of both methods are compared and the 0.7 used as cut-off value to determine the correlation, creating a network between those genes. Cytoscape was used to see the interaction of genes obtained with GGM with each other.

4. Results

The characteristics of the samples reported in the article are presented in Table-1. There is knowledge about the variability of distribution of data for each group with Boxplot (Figure-2 and Figure-3).

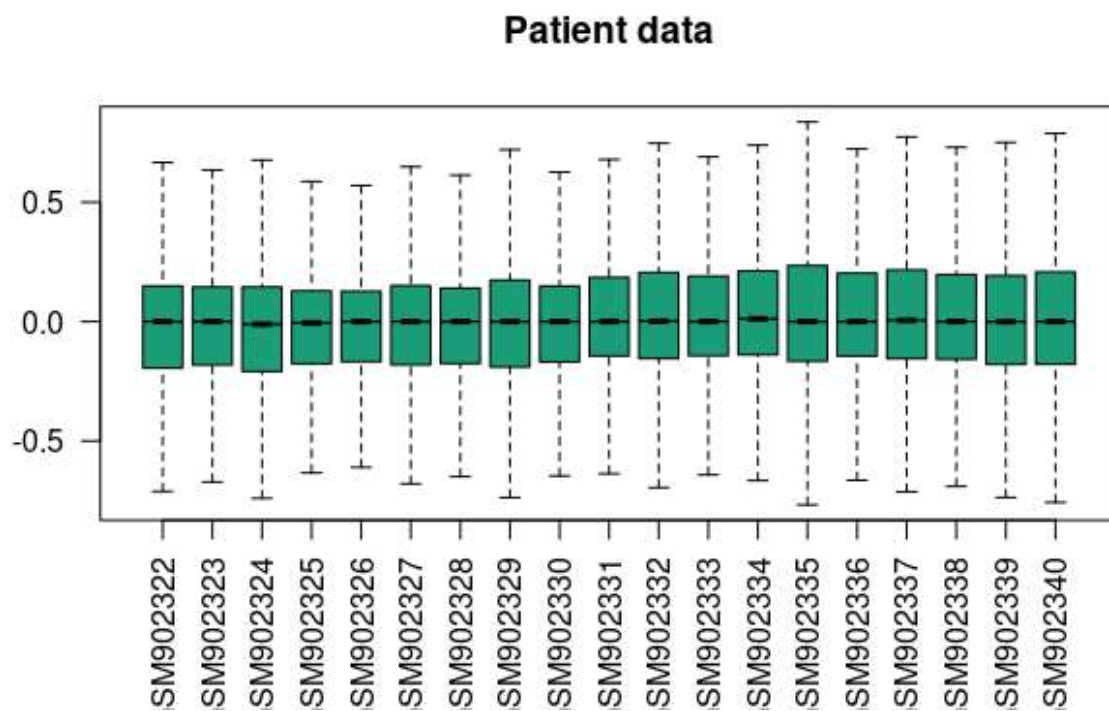


Figure 1 : Boxplot of Patients

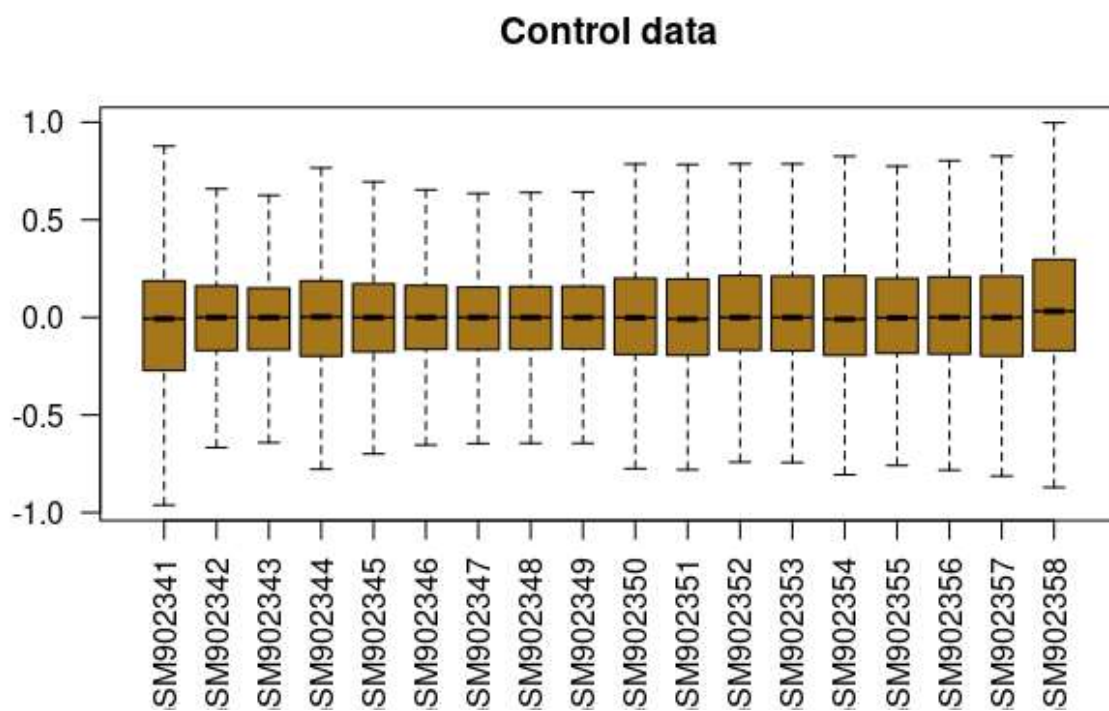


Figure 2 : Boxplot of Controls



218516_s_at		IMPAD1	1.49E-06
209808_x_at	inhibitor of growth family, member 1 [Source:HGNC Symbol;Acc:6062]	ING1	4.62E-07
210350_x_at	inhibitor of growth family, member 1 [Source:HGNC Symbol;Acc:6062]	ING1	3.92E-05
214692_s_at	jerky homolog (mouse) [Source:HGNC Symbol;Acc:6199]	JRK	3.98E-05
201473_at	jun B proto-oncogene [Source:HGNC Symbol;Acc:6205]	JUNB	6.60E-05
240592_at	ligand dependent nuclear receptor corepressor-like [Source:HGNC Symbol;Acc:30776]	LCORL	2.03E-05
222615_s_at		LOC100630923//LOC100289561//PRKRIP1	1.94E-05
203804_s_at	LUC7-like 3 (S. cerevisiae) [Source:HGNC Symbol;Acc:24309]	LUC7L3	6.28E-06
226385_s_at	mitochondrial assembly of ribosomal large subunit 1 [Source:HGNC Symbol;Acc:21721]	MALSU1	8.61E-05
242851_at		MFS4B	0.000107
231990_at		MIR6125//USP15	1.60E-05
220526_s_at	mitochondrial ribosomal protein L20 [Source:HGNC Symbol;Acc:14478]	MRPL20	0.000116
206491_s_at	N-ethylmaleimide-sensitive factor attachment protein, alpha [Source:HGNC Symbol;Acc:7641]	NAPA	1.64E-06
218455_at	NFS1 cysteine desulfurase [Source:HGNC Symbol;Acc:15910]	NFS1	3.75E-05
213773_x_at	NOP2/Sun domain family, member 5 [Source:HGNC Symbol;Acc:16385]	NSUN5	7.51E-05
213461_at	nudix (nucleoside diphosphate linked moiety X)-type motif 21 [Source:HGNC Symbol;Acc:13870]	NUDT21	7.78E-05
238856_s_at	pantothenate kinase 2 [Source:HGNC Symbol;Acc:15894]	PANK2	1.95E-05
201120_s_at	progesterone receptor membrane component 1 [Source:HGNC Symbol;Acc:16090]	PGRMC1	9.37E-05
211924_s_at	plasminogen activator, urokinase receptor [Source:HGNC Symbol;Acc:9053]	PLAUR	4.58E-05
213496_at		PLPPR4	6.82E-05
204285_s_at	phorbol-12-myristate-13-acetate-induced protein 1 [Source:HGNC Symbol;Acc:9108]	PMAIP1	6.53E-05
230777_s_at	PR domain containing 15 [Source:HGNC Symbol;Acc:13999]	PRDM15	2.42E-05
228590_at	pentatricopeptide repeat domain 3 [Source:HGNC Symbol;Acc:24717]	PTCD3	9.01E-05
202958_at	protein tyrosine phosphatase, non-receptor type 9 [Source:HGNC Symbol;Acc:9661]	PTPN9	6.11E-05
218360_at	RAB22A, member RAS oncogene family [Source:HGNC Symbol;Acc:9764]	RAB22A	1.87E-06
201395_at	RNA binding motif protein 5 [Source:HGNC Symbol;Acc:9902]	RBM5	5.49E-06
223169_s_at	ras homolog family member U [Source:HGNC Symbol;Acc:17794]	RHOU	2.42E-05
203843_at	ribosomal protein S6 kinase, 90kDa, polypeptide 3 [Source:HGNC Symbol;Acc:10432]	RPS6KA3	3.99E-05
1569599_at	SAM domain, SH3 domain and nuclear localization signals 1 [Source:HGNC	SAMSN1	4.13E-05

	Symbol;Acc:10528]		
209880_s_at	selectin P ligand [Source:HGNC Symbol;Acc:10722]	SELPLG	1.22E-06
214544_s_at	synaptosomal-associated protein, 23kDa [Source:HGNC Symbol;Acc:11131]	SNAP23	0.000109
206359_at	suppressor of cytokine signaling 3 [Source:HGNC Symbol;Acc:19391]	SOCS3	9.52E-06
205214_at	serine/threonine kinase 17b [Source:HGNC Symbol;Acc:11396]	STK17B	2.35E-05
1552542_s_at	T-cell activation RhoGTPase activating protein [Source:HGNC Symbol;Acc:15669]	TAGAP	6.78E-05
218268_at	TBC1 domain family, member 15 [Source:HGNC Symbol;Acc:25694]	TBC1D15	3.27E-05
219169_s_at	transcription factor B1, mitochondrial [Source:HGNC Symbol;Acc:17037]	TFB1M	1.34E-05
238635_at		TMEM267	5.30E-08
203869_at	ubiquitin specific peptidase 46 [Source:HGNC Symbol;Acc:20075]	USP46	8.54E-05
203959_s_at	zinc finger and BTB domain containing 40 [Source:HGNC Symbol;Acc:29045]	ZBTB40	1.02E-05
201531_at	ZFP36 ring finger protein [Source:HGNC Symbol;Acc:12862]	ZFP36	2.68E-05
225218_at	zinc finger, FYVE domain containing 27 [Source:HGNC Symbol;Acc:26559]	ZFYVE27	0.000108
203248_at	zinc finger protein 24 [Source:HGNC Symbol;Acc:13032]	ZNF24	7.07E-06
242463_x_at	zinc finger protein 600 [Source:HGNC Symbol;Acc:30951]	ZNF600	7.82E-05
1552499_a_at	zinc finger and SCAN domain containing 20 [Source:HGNC Symbol;Acc:13093]	ZSCAN20	2.15E-05
1559054_a_at	NA	NA	2.53E-05
229366_at	NA	NA	5.20E-06
237317_at	NA	NA	0.000115
238146_at	NA	NA	2.47E-06
241818_at	NA	NA	8.70E-05
241928_at	NA	NA	2.45E-05

*approximate p-values

Table-2: Significantly changed gene list for in peripheral arterial disease

We show our result of GO Analysis to the significantly changed genes for biological processes at Table-3. There are determined intersections cut off ≥ 10 . Also showing for KEGG pathway analysis at Table-4.

Term_name	Term_id	Intersection Size	Intersections
heterocycle metabolic process	GO:0046483	16	CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20

organic cyclic compound metabolic process	GO:1901360	16	CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular aromatic compound metabolic process	GO:0006725	15	CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NSUN5,PANK2,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
RNA metabolic process	GO:0016070	13	CREBRF,GCN1,ING1,LCORL,LUC7L3,NSUN5,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
nucleobase-containing compound metabolic process	GO:0006139	15	CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NSUN5,PANK2,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
nucleic acid metabolic process	GO:0090304	13	CREBRF,GCN1,ING1,LCORL,LUC7L3,NSUN5,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
heterocycle biosynthetic process	GO:0018130	11	CREBRF,GCN1,ING1,LCORL,NFS1,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular nitrogen compound metabolic process	GO:0034641	15	CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NSUN5,PANK2,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
primary metabolic process	GO:0044238	21	BEX4,CREBRF,GALK2,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PLPPR4,PRDM15,RBM5,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of primary metabolic process	GO:0080090	14	BEX4,CREBRF,GCN1,ING1,LCORL,NSUN5,PANK2,PRDM15,RBM5,SOC3,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of RNA metabolic process	GO:0051252	10	CREBRF,GCN1,ING1,LCORL,PRDM15,RBM5,ZBTB40,ZNF24,ZNF500,ZSCAN20
organic cyclic compound biosynthetic process	GO:1901362	11	CREBRF,GCN1,ING1,LCORL,NFS1,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of cellular metabolic process	GO:0031323	14	BEX4,CREBRF,GCN1,ING1,LCORL,NSUN5,PANK2,PRDM15,RBM5,SOC3,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of cellular macromolecule biosynthetic process	GO:2000112	10	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular metabolic process	GO:0044237	21	BEX4,CREBRF,GALK2,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PLPPR4,PRDM15,RBM5,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of macromolecule biosynthetic process	GO:0010556	10	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
nucleobase-containing compound biosynthetic process	GO:0034654	10	CREBRF,GCN1,ING1,LCORL,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of nitrogen compound metabolic process	GO:0051171	13	BEX4,CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,RBM5,SOC3,ZBTB40,ZNF24,ZNF500,ZSCAN20

regulation of nucleobase-containing compound metabolic process	GO:0019219	10	CREBRF,GCN1,ING1,LCORL,PRDM15,RBM5,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of cellular biosynthetic process	GO:0031326	10	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
aromatic compound biosynthetic process	GO:0019438	10	CREBRF,GCN1,ING1,LCORL,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of biosynthetic process	GO:0009889	10	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular nitrogen compound biosynthetic process	GO:0044271	11	CREBRF,GCN1,ING1,LCORL,NSUN5,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
nitrogen compound metabolic process	GO:0006807	19	BEX4,CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PRDM15,RBM5,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
organic substance metabolic process	GO:0071704	21	BEX4,CREBRF,GALK2,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PLPPR4,PRDM15,RBM5,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
gene expression	GO:0010467	13	CREBRF,GCN1,ING1,LCORL,LUC7L3,NSUN5,PRDM15,RBM5,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of gene expression	GO:0010468	11	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,RBM5,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular macromolecule metabolic process	GO:0044260	16	BEX4,CREBRF,GCN1,IMPAD1,ING1,LCORL,NFS1,NSUN5,PRDM15,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of metabolic process	GO:0019222	14	BEX4,CREBRF,GCN1,ING1,LCORL,NSUN5,PANK2,PRDM15,RBM5,SOC3,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of macromolecule metabolic process	GO:0060255	13	BEX4,CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,RBM5,SOC3,ZBTB40,ZNF24,ZNF500,ZSCAN20
macromolecule metabolic process	GO:0043170	18	BEX4,CREBRF,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PRDM15,RBM5,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular biosynthetic process	GO:0044249	12	CREBRF,GCN1,ING1,LCORL,NFS1,NSUN5,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
metabolic process	GO:0008152	21	BEX4,CREBRF,GALK2,GCN1,IMPAD1,ING1,LCORL,LUC7L3,NFS1,NSUN5,PANK2,PLPPR4,PRDM15,RBM5,SOC3,STK17B,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular macromolecule biosynthetic process	GO:0034645	10	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
cellular process	GO:0009987	28	BEX4,COA5,CREBRF,GALK2,GCN1,IMPAD1,ING1,JRK,LCORL,LUC7L3,NAPA,NFS1,NSUN5,PANK2,PLPPR4,PRDM15,RAB22A,RBM5,RHOU,SELPLG,SOC3,STK17B,TAGAP,TFB1M,ZBTB40,ZNF24,ZNF500,ZSCAN20

organic substance biosynthetic process	GO:1901576		12	CREBRF,GCN1,ING1,LCORL,NFS1,NSUN5,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
biosynthetic process	GO:0009058		12	CREBRF,GCN1,ING1,LCORL,NFS1,NSUN5,PANK2,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
macromolecule biosynthetic process	GO:0009059		10	CREBRF,GCN1,ING1,LCORL,NSUN5,PRDM15,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of cellular process	GO:0050794		20	BEX4,CREBRF,GCN1,ING1,JRK,LCORL,NAPA,NSUN5,PANK2,PLPPR4,PRDM15,RBM5,RHO,SOCS3,STK17B,TAGAP,ZBTB40,ZNF24,ZNF500,ZSCAN20
positive regulation of cellular process	GO:0048522		11	CREBRF,GCN1,ING1,JRK,NSUN5,PRDM15,RBM5,RHO,SOCS3,STK17B,ZNF24
cellular component organization or biogenesis	GO:0071840		12	COA5,ING1,LUC7L3,NAPA,NFS1,NSUN5,PANK2,PLPPR4,RAB22A,RBM5,RHO,TFB1M
anatomical structure development	GO:0048856		11	IMPAD1,NAPA,NSUN5,PANK2,PLPPR4,PRDM15,RHO,SELPLG,SOCS3,ZBTB40,ZNF24
biological regulation	GO:0065007		21	BEX4,CREBRF,GCN1,ING1,JRK,LCORL,NAPA,NSUN5,PANK2,PLPPR4,PRDM15,RAB22A,RBM5,RHO,SOCS3,STK17B,TAGAP,ZBTB40,ZNF24,ZNF500,ZSCAN20
regulation of biological process	GO:0050789		20	BEX4,CREBRF,GCN1,ING1,JRK,LCORL,NAPA,NSUN5,PANK2,PLPPR4,PRDM15,RBM5,RHO,SOCS3,STK17B,TAGAP,ZBTB40,ZNF24,ZNF500,ZSCAN20
positive regulation of biological process	GO:0048518		11	CREBRF,GCN1,ING1,JRK,NSUN5,PRDM15,RBM5,RHO,SOCS3,STK17B,ZNF24
cellular component organization	GO:0016043		11	COA5,ING1,LUC7L3,NAPA,NFS1,PANK2,PLPPR4,RAB22A,RBM5,RHO,TFB1M
developmental process	GO:0032502		11	IMPAD1,NAPA,NSUN5,PANK2,PLPPR4,PRDM15,RHO,SELPLG,SOCS3,ZBTB40,ZNF24
cell communication	GO:0007154		10	CREBRF,GCN1,JRK,NAPA,PLPPR4,PRDM15,RHO,SOCS3,STK17B,TAGAP
cellular response to stimulus	GO:0051716		11	CREBRF,GCN1,JRK,PLPPR4,PRDM15,RHO,SELPLG,SOCS3,STK17B,TAGAP,ZBTB40
multicellular organismal process	GO:0032501		11	CREBRF,IMPAD1,NAPA,NSUN5,PANK2,PLPPR4,PRDM15,SELPLG,SOCS3,ZBTB40,ZNF24
response to stimulus	GO:0050896		11	CREBRF,GCN1,JRK,PLPPR4,PRDM15,RHO,SELPLG,SOCS3,STK17B,TAGAP,ZBTB40

Table-3: GO Analysis to the significantly changed genes for biological processes

Source	Term_name	Term_id	adjusted_p_value	effective_domain_size	intersections
KEGG	Sulfur relay system	KEGG:04122	0.0897916257	7987	NFS1
KEGG	Biosynthesis of cofactors	KEGG:01240	0.0897916257	7987	NFS1,PANK2
KEGG	Sulfur metabolism	KEGG:00920	0.0897916257	7987	IMPAD1

KEGG	Metabolic pathways	KEGG:01100	0.2026230009	7987	IMPAD1,NFS1,PAN K2
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Table-4: KEGG Pathway enrichment analysis results

The GGM based network inference and the Pearson correlation methods used to determine the correlation between 6 genes which are most significantly changed genes. The results of the method are given in Table 5. To compare these two methods; there are no differences between results. We believe that the margin of error is reduced since the pval value of the genes are very low. Therefore, the output of the methods became the same. Also visualized interactions between these genes from Figure-5.

The correlated genes: *ING1*, *GCN1*, *RAB22A*, *IMPAD1* and *TMEM267*.

a) GGM						
	ING1	SELPLG	GCN1	RAB22A	IMPAD1	TMEM267
ING1	-1	-0.5641594505	0.642269882	-0.7330851432	-0.622925251	0.6216073443
SELPLG	-0.5641594505	-1	0.6042985582	-0.6124746153	-0.535381359	0.6880609419
GCN1	0.642269882	0.6042985582	-1	0.6604092974	0.6333288467	-0.7750432204
RAB22A	-0.7330851432	-0.6124746153	0.6604092974	-1	-0.7135595172	0.6136073953
IMPAD1	-0.622925251	-0.535381359	0.6333288467	-0.7135595172	-1	0.6687173492
TMEM267	0.6216073443	0.6880609419	-0.7750432204	0.6136073953	0.6687173492	-1
b) Pearson						
	ING1	SELPLG	GCN1	RAB22A	IMPAD1	TMEM267
ING1	1	0.5641594505	-0.642269882	0.7330851432	0.622925251	-0.6216073443
SELPLG	0.5641594505	1	-0.6042985582	0.6124746153	0.535381359	-0.6880609419
GCN1	-0.642269882	-0.6042985582	1	-0.6604092974	-0.6333288467	0.7750432204
RAB22A	0.7330851432	0.6124746153	-0.6604092974	1	0.7135595172	-0.6136073953
IMPAD1	0.622925251	0.535381359	-0.6333288467	0.7135595172	1	-0.6687173492
TMEM267	-0.6216073443	-0.6880609419	0.7750432204	-0.6136073953	-0.6687173492	1
c) > 0.7 for cutoff value						
	ING1	SELPLG	GCN1	RAB22A	IMPAD1	TMEM267

ING1	1	0	0	1	0	0
SELPLG	0	1	0	0	0	0
GCN1	0	0	1	0	0	1
RAB22A	1	0	0	1	1	0
IMPAD1	0	0	0	1	1	0
TMEM267	0	0	1	0	0	1

Table-5 : Comparing GGM and Pearson values for determining interaction in genes :

a) GGM, b) Pearson, c) Result of cutoff value

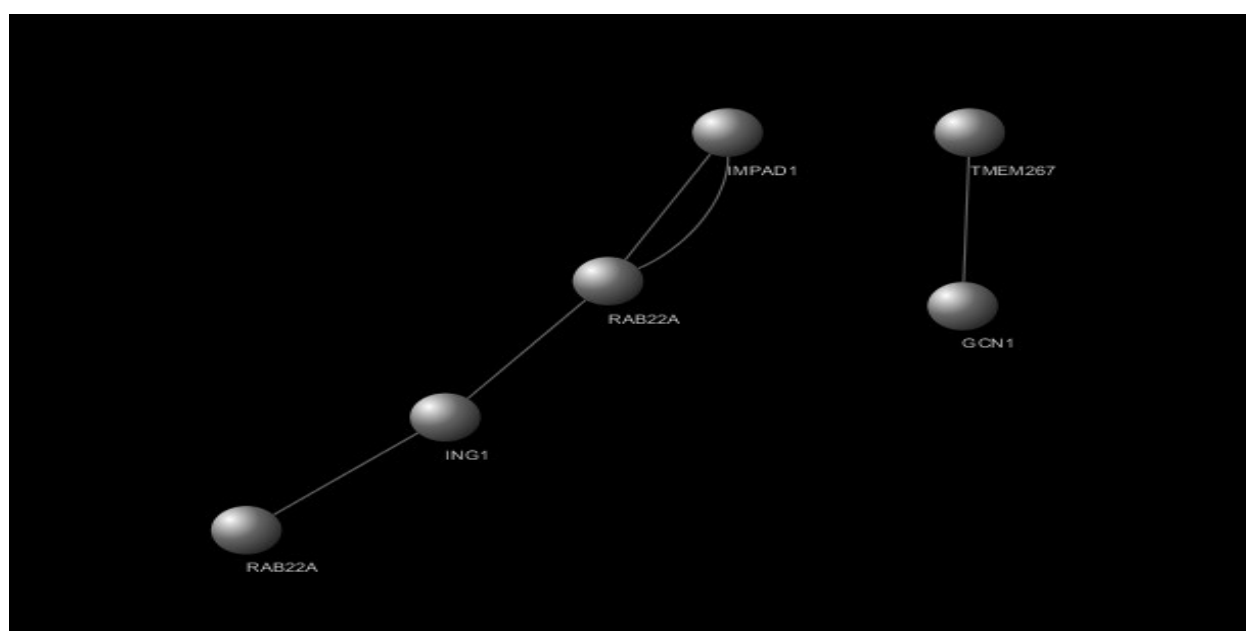


Figure-5 : Visualisation of interaction between genes by GGM result

source	term_name	term_id	intersection_size	intersections
GO:BP	nucleobase-containing compound metabolic process	GO:0006139	3	ING1,GCN1,IMPAD1
GO:BP	cellular aromatic compound metabolic process	GO:0006725	3	ING1,GCN1,IMPAD1
GO:BP	heterocycle metabolic process	GO:0046483	3	ING1,GCN1,IMPAD1
GO:BP	organic cyclic compound metabolic process	GO:1901360	3	ING1,GCN1,IMPAD1
GO:BP	cellular nitrogen compound metabolic process	GO:0034641	3	ING1,GCN1,IMPAD1
GO:BP	cellular macromolecule metabolic process	GO:0044260	3	ING1,GCN1,IMPAD1
GO:BP	macromolecule metabolic process	GO:0043170	3	ING1,GCN1,IMPAD1

GO:BP	nitrogen compound metabolic process	GO:0006807	3	ING1,GCN1,IMPAD1
GO:BP	primary metabolic process	GO:0044238	3	ING1,GCN1,IMPAD1
GO:BP	cellular metabolic process	GO:0044237	3	ING1,GCN1,IMPAD1
GO:BP	organic substance metabolic process	GO:0071704	3	ING1,GCN1,IMPAD1
GO:BP	metabolic process	GO:0008152	3	ING1,GCN1,IMPAD1
GO:BP	biological regulation	GO:0065007	3	ING1,GCN1,RAB22A
GO:BP	cellular process	GO:0009987	4	ING1,GCN1,RAB22A,IMPAD1

a) Biological process

source	term_name	term_id	intersection_size	Effective domain_size	intersections
GO:MF	ion binding	GO:0043167	3	18694	ING1,RAB22A,IMPAD1
GO:MF	binding	GO:0005488	5	18694	ING1,GCN1,RAB22A,IMPAD1,TMEM267
GO:MF	protein binding	GO:0005515	4	18694	ING1,GCN1,RAB22A,TMEM267

b) Molecular Function

Table-6: GO Analysis to common genes by GGM result for biological processes and molecular function

5. Discussions

We report for gene expression analysis of PBMC to identify genes differentially expressed in patients with PAD.

Firstly, we control the dataset for the variability of distribution of data for each group. According to Boxplot, there is no completely different distribution in whole data. The next step is to identify the different samples within the group. In Figure-3, it is clear which sample should not be evaluated within the group. One of the sample IDs is GSM902323 that is specified as a patient sample in the data but according to both PCA, dendrogram and heap map plot, it acts as a control sample. Therefore, excluded from assessment. When we do the t-test with this example, the statistical significance gene list is 36. When our analysis and statistical evaluations except this sample, we get 64 significance genes.

Secondly, after determining significantly different genes, is 64 genes, we use these genes to the interpretation of the function or pathways associated with the disease. We aim to get a result for molecular clues. When examined to the enrichment analysis results of intersections of these genes, there are the majority biosynthetic process, metabolic process, cellular process, structure developmental, etc as a biological process. Also, three important pathways we take from KEGG are important in the metabolic process.

As a result, GO annotations and pathway enrichment using KEGG provide an understanding of the function and processes associated with disease by the significance change genes.

GGM based inference gives us 5 related genes in case of the cut-off 0.7. Since this cut-off is not reliable enough to give accurate relations and interactions g: profiler results are strongly affected. There is no significant correlation or similarity between these genes, indeed. Table-6 in the result part strongly supports our deduction. Masud et.al^[1] identified some other genes which are strongly relevant with PAD pathophysiology. Those genes are commonly responsible for “response to wounding”, “leukocyte activation”, “apoptosis” etc.

For biological processes; those genes are expressed together in very common pathways such as “nucleobase containing compound metabolic process” or “heterocycle (Table-6a) compound metabolic process”. Also, in molecular function, genes are involved together in very common functions such as protein binding or ion binding functions (Table-6b). So we can easily say that those genes are not expressed together.

To conclusion, there is no accordance between our GGM results and literature or in g: profiler. In our opinion the mismatch arises from our lack of statistical analysis such that determining the up or down regulated genes and lack of clustering skills.

6. References

1. Masud R., Shameer K, Dhar A.,Ding K., J Kullo I :Gene expression profiling of peripheral blood mononuclear cells in the setting of peripheral arterial disease. J Clin Bioinforma. 2012; 2: 6. Published online 2012 Mar 12. doi: 10.1186/2043-9113-2-6
2. Smedley D, Haider S, Ballester B, Holland R, London D, Thorisson G, Kasprzyk A: BioMart-biological queries made easy. BMC Genomics 2009, 10:22.
3. The NCBI the Gene Expression Omnibus (GEO) database
<http://www.ncbi.nlm.nih.gov/geo/>
4. Swedish Council on Health Technology Assessment. Peripheral Arterial Disease – Diagnosis and Treatment: A Systematic Review [Internet]. Stockholm: Swedish Council on Health Technology Assessment (SBU); 2008 Nov. SBU Yellow Report No. 187. PMID: 28876730.